	Value of x increases.
3(a)	Period of satellite = 24 h The plane of orbit of the satellite is in the plane of the equator of the Earth.
3(b)	Although the satellite's speed is constant, its velocity is changing due to the change in its direction. Since acceleration is the rate of change of velocity, it has an acceleration towards the centre of the orbit. The direction of the gravitational force on the satellite is towards the centre of the orbit. This direction is perpendicular to the direction of its velocity.
3(c)(i)	Gravitational force on satellite provides the centripetal force on satellite. Gravitational force $=\dfrac{GMm}{r^2}$ where r is the distance between satellite and Earth Centripetal force $=\dfrac{mv^2}{r}$ where r is the radius of the orbit $\dfrac{GMm}{r^2}=\dfrac{mv^2}{r}$

No.	Solution
3(c)(ii)	$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\left(\frac{2\pi}{T}\right)^2$ $\frac{GM}{r^3} = \frac{4\pi^2}{T^2}$ $r^3 = \frac{GMT^2}{4\pi^2} = \frac{6.67\times10^{-11}\left(5.97\times10^{24}\right)\left(24\times60\times60\right)^2}{4\pi^2}$ $r = 4.2227\times10^7$ m $h + 6.37\times10^6 = 4.2227\times10^7$ $h = 3.59\times10^7$ m
4(a)	Ω